



Mata Kuliah Pemograman Visual (Desktop)



Materi Pertemuan ke-10

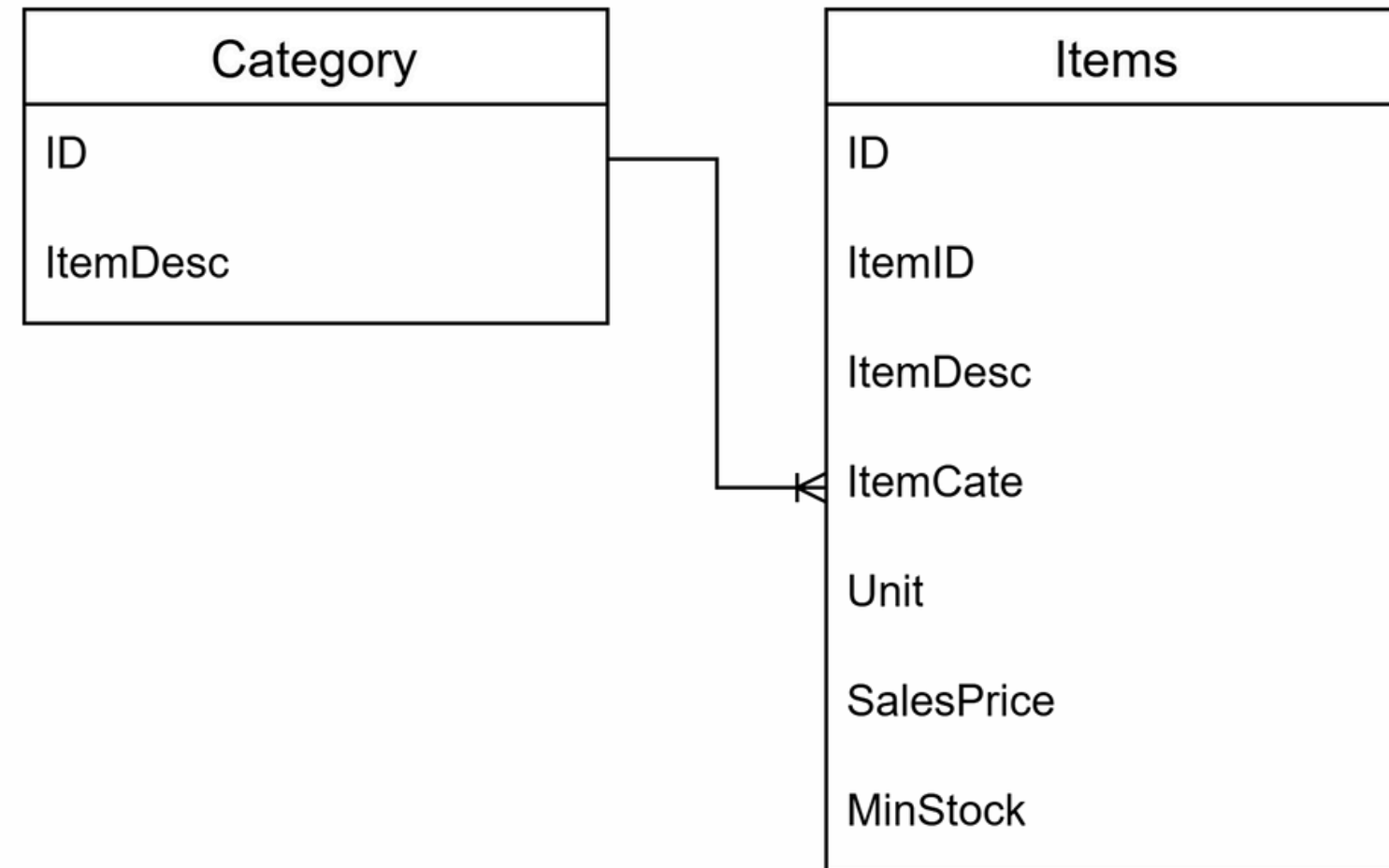
Penerapan MVC pada Akses Database Berelasi di VB.NET

Dosen Pengampu Asep Muhidin, S.Kom., M.Kom.

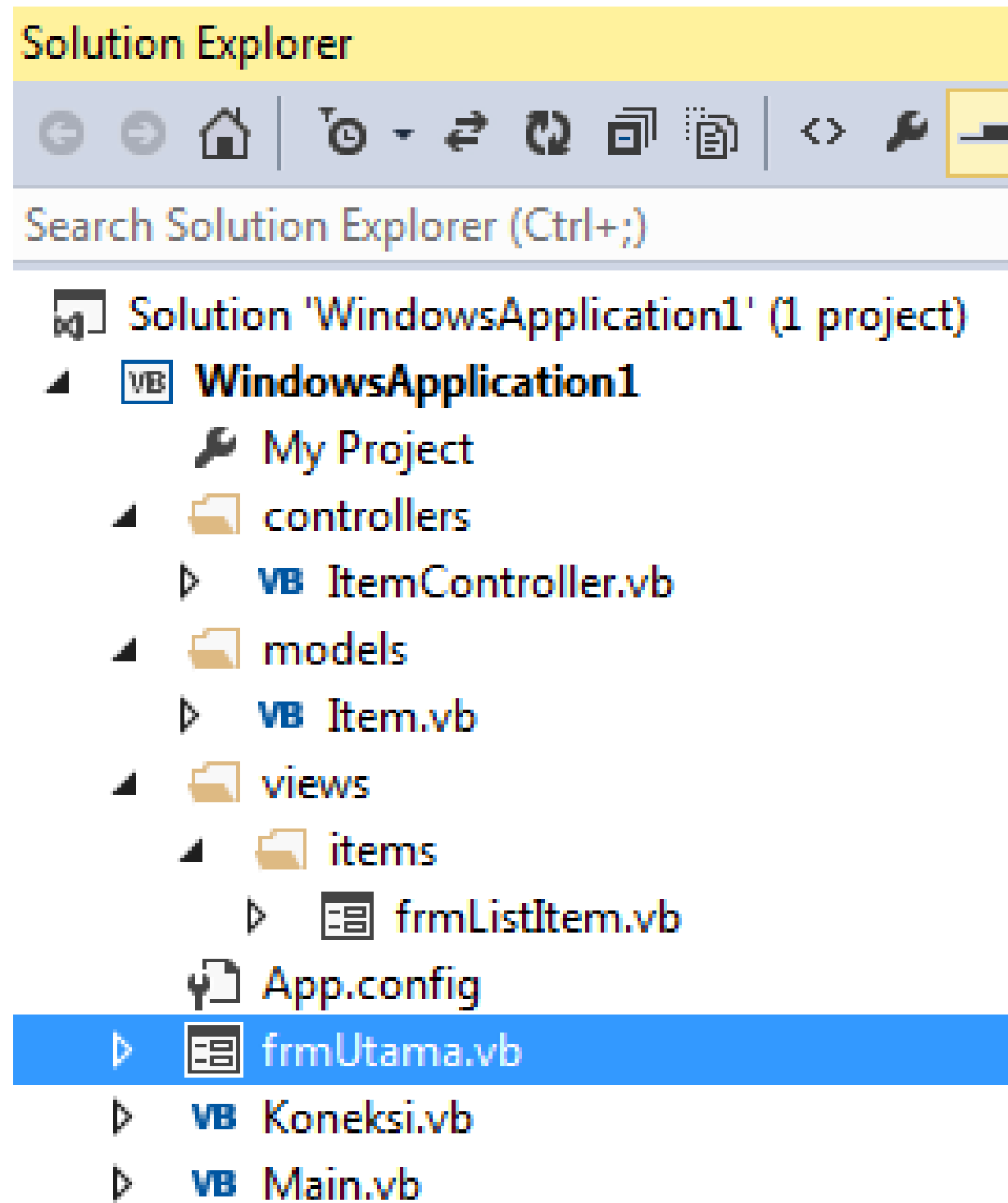




Struktur Database



Struktur Project

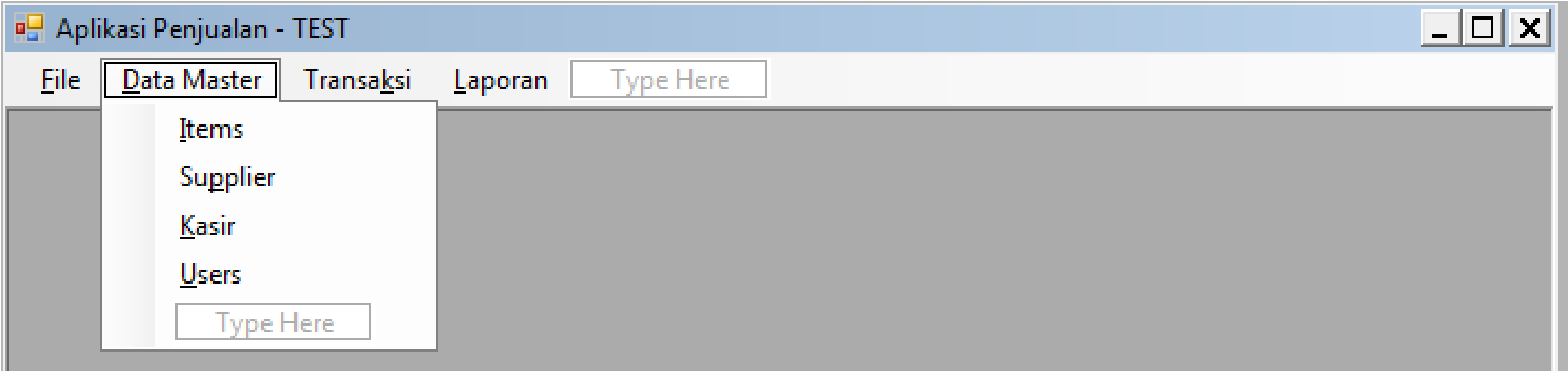


Main.vb



```
Module Main
    Public Sub main()
        ' Panggil form utama
        Application.Run(New frmUtama())
    End Sub
End Module
```

frmUtama.frm



Public Class frmUtama

```
Private Sub KeluarToolStripMenuItem_Click(sender As Object, e As EventArgs) Handles  
KeluarToolStripMenuItem.Click  
Application.Exit()  
End Sub
```

```
Private Sub ItemsToolStripMenuItem_Click(sender As Object, e As EventArgs) Handles  
ItemsToolStripMenuItem.Click  
Dim frm As New frmListItem  
  
frm.MdiParent = Me  
frm.Show()  
End Sub  
End Class
```

+	Icon	(Icon)
	IsMdiContainer	<u>True</u>
	MainMenuStrip	MenuStrip1

IsMdiContainer : True

Koneksi.vb



```
Imports MySql.Data.MySqlClient
```

```
Module Koneksi
```

```
    Private ReadOnly server As String = "192.168.100.94"
```

```
    Private ReadOnly database As String = "dbpenjualan"
```

```
    Private ReadOnly username As String = "root"
```

```
    Private ReadOnly password As String = ""
```

```
    Private conn As MySqlConnection
```

```
    ' Fungsi utama untuk membuka koneksi
```

```
    Public Function OpenConnection() As MySqlConnection
```

```
        Try
```

```
            If conn Is Nothing Then
```

```
                Dim connString As String = String.Format("server={0};user id={1};password={2};database={3};", server,  
                    username, password, database)
```

```
                conn = New MySqlConnection(connString)
```

```
            End If
```

```
            If conn.State = ConnectionState.Closed OrElse conn.State = ConnectionState.Broken Then
```

```
                conn.Open()
```

```
            End If
```

```
            Return conn
```

```
        Catch ex As MySqlException
```

```
            MessageBox.Show("Koneksi gagal: " & ex.Message, "Error Koneksi", MessageBoxButtons.OK, MessageBoxIcon.Error)
```

```
            Return Nothing
```

```
        End Try
```

```
    End Function
```


Koneksi.vb



```
' Menutup koneksi
Public Sub CloseConnection()
    Try
        If conn IsNot Nothing AndAlso conn.State = ConnectionState.Open Then
            conn.Close()
        End If
    Catch ex As Exception
        MessageBox.Show("Gagal menutup koneksi: " & ex.Message, "Error", MessageBoxButtons.OK, MessageBoxIcon.Error)
    Finally
        conn = Nothing
    End Try
End Sub

' Mengecek status koneksi
Public Function IsConnected() As Boolean
    Try
        Return conn IsNot Nothing AndAlso conn.State = ConnectionState.Open
    Catch ex As Exception
        Return False
    End Try
End Function
End Module
```

ItemModel.vb

```
Imports MySql.Data.MySqlClient
```

```
Public Class ItemModel
```

```
    Public Property id As Integer
```

```
    Public Property itemID As String
```

```
    Public Property itemDesc As String
```

```
    Public Property itemCate As Integer
```

```
    Public Property unit As String
```

```
    Public Property salesPrice As Integer
```

```
    Public Property minStock As Integer
```

```
    Public Shared Function getAll() As DataTable
```

```
        Dim dt As New DataTable
```

```
        Dim query As String = "SELECT items.*,categoryDesc " & _  
        " FROM items LEFT JOIN category ON items.itemCate =  
        category.id "
```

```
        Using conn = Koneksi.OpenConnection()
```

```
            Using cmd As New MySqlCommand(query, conn)
```

```
                dt.Load(cmd.ExecuteReader())
```

```
            End Using
```

```
        End Using
```

```
        Return dt
```

```
    End Function
```

```
End Class
```



Using conn adalah blok Using di VB.NET yang berfungsi untuk mengelola resource otomatis, khususnya untuk objek yang harus ditutup setelah dipakai.

Using ... End Using berarti:

1. Membuka koneksi
2. Menjalankan semua perintah di dalam blok
3. Setelah selesai (atau error), koneksi akan ditutup otomatis dan memory dibersihkan.

ItemController.vb

```
Public Class ItemController
    Public Function LoadItems() As DataTable
        Return ItemModel.getAll()
    End Function
End Class
```

frmListItem.frm



frmListItem.vb [Design]*

Form Daftar Items

Search

	ItemID	Item Description	Item Category	Unit	Sales Price	Min. Stock
•						

frmListItem.frm



```
Public Class frmListItem
    Dim controller As New ItemController
    Private Sub frmListItem_Load(sender As Object, e As EventArgs) Handles MyBase.Load
        LoadGrid()
    End Sub
    Sub LoadGrid()
        DataGridView1.AutoGenerateColumns = False
        Dim dt As DataTable = controller.LoadItems()

        DataGridView1.DataSource = dt
        DataGridView1.Columns("salesPrice").HeaderCell.Style.Alignment =
DataGridViewContentAlignment.MiddleRight

    End Sub

    Private Sub txtSearch_TextChanged(sender As Object, e As EventArgs) Handles txtSearch.TextChanged
        Dim dv As DataView = CType(DataGridView1.DataSource, DataTable).DefaultView
        dv.RowFilter = "itemDesc LIKE '%" & txtSearch.Text & "%' OR categoryDesc LIKE '%" &
txtSearch.Text & "%'"
    End Sub
End Class
```